Selecting the appropriate surfactant and surface modification process for a UHMWPE-matrix nano Cu/Graphene composite is critical because UHMWPE is highly non-polar and chemically inert, while copper nanoparticles and graphene nanoplatelets (GNPs) are inherently incompatible with this hydrophobic matrix . The core challenge is bridging this polarity mismatch to achieve stable, uniform dispersion without compromising the intrinsic properties of the fillers.

□ Surfactant Selection Criteria

The choice of surfactant or dispersing agent depends on the filler type and the processing route. Two primary strategies exist: **non-covalent physical adsorption** using surfactants, and **covalent surface functionalization** using coupling agents.

For **graphene nanoplatelets (GNPs)**, common surfactants include **polyvinyl pyrrolidone (PVP)**, **sodium dodecyl benzene sulfonate (SDBS)**, and **octyl phenol ethoxylate**. These work by adsorbing onto the graphene surface and providing steric or electrostatic repulsion to prevent re-agglomeration driven by Van der Waals forces . However, a significant drawback is that these surfactants can interfere with the nanomaterial's mechanical, electrical, and thermal properties, and some, like PVP, are difficult to remove completely .

For **copper nanoparticles and nanowires**, the surfactant must address both oxidation susceptibility and the hydrophilic nature of the copper surface. A highly effective approach uses **oleylamine** as both a synthesis ligand and a dispersing agent. When copper nanowires are synthesized in a mixture of oleylamine and octadecene, the amine group coordinates to the copper surface, while the long alkyl chain renders the particles **apolar and dispersible in organic solvents** like hexane or 1,2-dichlorobenzene . This oleylamine layer is crucial; without it, the nanowires tend to agglomerate severely in polyethylene. The oleylamine modification effectively converts the hydrophilic copper surface into one compatible with the hydrophobic polyethylene matrix .

□ Surface Modification Processes

Oleylamine Coating for Copper (In-Situ Synthesis): The most elegant method for copper nanowires is to perform the synthesis directly in a coordinating solvent. Copper(II)

acetylacetonate and copper(II) chloride are reacted in a mixture of **oleylamine and 1-octadecene** at elevated temperatures (120°C then 180°C) . The resulting copper nanowires are inherently coated with an oleylamine layer. The critical process detail is to use **fresh oleylamine** stored under inert conditions; oxidized oleylamine leads to poorly dispersible products . After synthesis, the nanowires are washed with hexane and can be redispersed in solvents like 1,2-dichlorobenzene for subsequent composite preparation .

Silane Coupling Agent Treatment for Copper Nanoparticles: For copper nanoparticles or microparticles, a more robust covalent modification can be achieved using **silane coupling agents**. A representative process involves :

1. **Pre-treatment**: The copper powder is ultrasonically dispersed in ethanol, filtered, and dried.

2. **Surface Activation**: The powder is etched in **10 wt% HCl** for 60–90 seconds to strip the native oxide layer and activate the surface, followed by rinsing to neutral pH and vacuum drying.

3. **Silane Grafting**: The activated powder is added to a solution of **3-mercaptopropyltriethoxysilane (MPTES)** (5 wt% in ethanol/water 9:1, pH adjusted to 4.5–5.0 with acetic acid) and stirred at **50°C for 2 hours**.

4. **Purification**: The modified powder is washed with ethanol to remove unreacted silane and vacuum dried at 80°C then 100°C .

The **Hansen Solubility Parameter (HSP)** theory provides a systematic method for selecting the optimal silane coupling agent. The HSP of a copper surface shifts toward that of the treatment agent after modification, so choosing a silane whose HSP matches the UHMWPE matrix (or the processing solvent) can predictably improve dispersibility .

Graphene Surface Modification via Functionalization or Coating: For graphene, surface modification often involves using **functionalized graphene** (e.g., graphene oxide with hydroxyl, carboxyl, or epoxy groups) which disperses more readily in polar environments before being compounded . Alternatively, a **coating approach** can be used: UHMWPE is dip-coated in a **1 wt% GNP dispersion in ethanol** (prepared by ultrasonication at 180 W for 30 min), followed by baking at 80°C for 24 h. This creates a GNP-rich surface layer. **Electron beam irradiation (EBI)** can subsequently be applied to enhance GNP adhesion and induce crosslinking in the UHMWPE surface .

Polydopamine (PDA) and Reduced Graphene Oxide (RGO) Synergistic Coating: For UHMWPE fibers, a sequential modification using **KH550 pretreatment**, **dopamine oxidative self-polymerization** (PDA coating), and **RGO anchoring** creates a functionalized surface. The PDA layer strengthens metal adhesion, while RGO enhances conductivity, enabling subsequent **electroless copper deposition**. This route transforms the insulating UHMWPE surface into a conductive Cu/RGO coating .

□ Practical Selection Guidance

For a **UHMWPE-matrix composite with nano Cu and graphene**, the most direct strategy is:

- For **copper nanowires**: Synthesize them in the presence of **oleylamine** to obtain a dispersible, apolar surface layer .

- For **copper nanoparticles**: Use a **silane coupling agent** (selected via HSP theory) with acid pre-etching to covalently bond a compatible organic layer .

- For **graphene**: Use **functionalized graphene (GO/RGO)** or a **dip-coating/EBI** approach to create a robust, adhered surface layer .